

DAY 03 — Conditional & Criteria-Based Analysis

Excel Notes

1. Learning Objectives

- Use logical conditions to classify and evaluate business data.
- Combine multiple conditions using AND and OR.
- Handle formula errors using IFERROR.
- Summarize business data using SUMIF/SUMIFS, COUNTIF/COUNTIFS, and AVERAGEIF/AVERAGEIFS.
- Translate business rules into Excel formulas.
- Apply conditional and criteria-based formulas to a sales-performance scenario.

2. Day 03 Function Summary

Function	Purpose	Core Idea
IF	Make a decision based on one condition.	If TRUE → one result; otherwise → another.
Nested IF	Handle multiple business conditions.	Check conditions in sequence.
AND	Require all conditions to be TRUE.	All conditions must pass.
OR	Require at least one condition to be TRUE.	Any condition can pass.
IFERROR	Replace formula errors with a useful result.	If formula errors → return alternative.
SUMIF	Sum values using one criterion.	One condition → sum matching values.
SUMIFS	Sum values using multiple criteria.	Multiple conditions → sum matching values.
COUNTIF	Count records using one criterion.	One condition → count matches.
COUNTIFS	Count records using multiple criteria.	Multiple conditions → count matches.
AVERAGEIF	Average values using one criterion.	One condition → average matches.
AVERAGEIFS	Average values using multiple criteria.	Multiple conditions → average matches.

3. IF

Use IF when a business rule has two possible outcomes.

Syntax: =IF(condition, value_if_true, value_if_false)

Example:

=IF([@Sales]>=[@Target], "Achieved", "Not Achieved")

Key point: Use >= when a salesperson should be considered successful when Sales are equal to Target.

4. Nested IF

Nested IF is used when there are more than two possible outcomes.

Example classification:

- Sales $\geq$ 120% of Target → Excellent
- Sales $\geq$ 100% of Target → Achieved
- Sales $\geq$ 80% of Target → Needs Improvement
- Sales $<$ 80% of Target → Poor

Formula:

```
=IF(D9>=120%, "Excellent", IF(D9>=100%, "Achieved", IF(D9>=80%, "Needs_Improvement", "Poor")))
```

Important: The order of conditions matters. Check the strongest/highest condition first.

5. AND

AND returns TRUE only when every condition is TRUE.

Business example: Bonus eligibility requires Sales $\geq$ Target AND Rating $\geq$ 4.

```
=IF(AND([@Sales]>=[@Target],[@Rating]>=4), "Eligible_for_Bonus", "Not_Eligible")
```

Mental model: TRUE + TRUE = TRUE. If even one condition is FALSE, AND returns FALSE.

6. OR

OR returns TRUE when at least one condition is TRUE.

Business example: Priority follow-up when Sales $<$ Target OR Rating $<$ 3.

```
=IF(OR(I9<J9, K9<3), "Urgent_Followup", "Normal_Followup")
```

Mental model: Only one condition needs to be TRUE.

7. IFERROR

IFERROR prevents Excel errors such as #DIV/0! from appearing in the final report.

Example: Sales per Lead = Sales / Leads. If Leads = 0, show No_Lead.

```
=IFERROR([@Sales]/[@Leads], "No_Lead")
```

Common use: division calculations, lookup results, and calculations where missing/zero data can create errors.

8. SUMIF

SUMIF adds values that meet one condition.

Syntax: =SUMIF(criteria_range, criteria, sum_range)

Example: Total Kenya sales:

=SUMIF(A:A,E5,B:B)

Example result: If Kenya appears with sales of 20,000, 25,000 and 30,000, total Kenya sales = 75,000.

9. SUMIFS

SUMIFS adds values that meet multiple conditions.

Syntax: =SUMIFS(sum_range, criteria_range1, criteria1, criteria_range2, criteria2)

Example: Kenya + Loans:

=SUMIFS(K:K,I:I,"Kenya",J:J,"Loans")

Important: In SUMIFS, the sum_range comes first.

10. COUNTIF

COUNTIF counts records that meet one condition.

Syntax: =COUNTIF(range, criteria)

Example: Count salespeople who achieved target:

=COUNTIF(B:B,"Achieved")

11. COUNTIFS

COUNTIFS counts records that satisfy multiple conditions.

Syntax: =COUNTIFS(criteria_range1, criteria1, criteria_range2, criteria2, ...)

Example: Kenya + Loans + Achieved:

=COUNTIFS(I:I,"Kenya",J:J,"Loans",K:K,"Achieved")

12. AVERAGEIF

AVERAGEIF calculates the average of values meeting one condition.

Syntax: =AVERAGEIF(criteria_range, criteria, average_range)

Example: Average Kenya sales:

`=AVERAGEIF(G:G,J4,H:H)`

13. AVERAGEIFS

AVERAGEIFS calculates an average using multiple criteria.

Syntax: `=AVERAGEIFS(average_range, criteria_range1, criteria1, criteria_range2, criteria2)`

Example: Average Kenya + Loans sales:

`=AVERAGEIFS(K:K,I:I,M2,J:J,N2)`

Important: In AVERAGEIFS, the average_range comes first.

14. Structured References

During Day 03, Excel Table structured references were used, such as:

- `[@Sales]`
- `[@Target]`
- `[@Rating]`
- `[@Leads]`

These references make formulas easier to read and automatically work with rows in an Excel Table.

15. Business-Condition Calculations

The key skill is translating a business rule into Excel logic.

Example: A salesperson is eligible for a bonus only when all three conditions are met:

- Sales >= Target
- Rating >= 4
- Leads >= 50

`=IF(AND([@Sales]>=[@Target],[@Rating]>=4,[@Leads]>=50),"Eligible","Not Eligible")`

For the exercise, Salespersons A and E were eligible.

16. Sales Performance Exercise — Key Thinking

The final exercise required choosing the appropriate function based on the business question, without being told which function to use.

Business Question	Relevant Day 03 Skill
Total Sales — Kenya	SUMIF
Insurance Sales — Kenya	SUMIFS
Count Achieved — Kenya	COUNTIFS
Average Sales — Kenya + Insurance	AVERAGEIFS

Bonus Eligibility	IF + AND
Highest Sales Country	Conditional logic was practiced using IF based on already-calculated totals.

17. Key Differences to Remember

Function	Conditions	Action/Logic
SUMIF	1 condition	Sum
SUMIFS	Multiple conditions	Sum
COUNTIF	1 condition	Count
COUNTIFS	Multiple conditions	Count
AVERAGEIF	1 condition	Average
AVERAGEIFS	Multiple conditions	Average
AND	All conditions	TRUE only if all pass
OR	Any condition	TRUE if at least one passes

18. Day 03 Takeaways

- Do not start with the function; start with the business question.
- IF converts a business rule into a decision.
- AND means all conditions must be true; OR means at least one condition can be true.
- SUMIF/COUNTIF/AVERAGEIF handle one criterion; their S versions handle multiple criteria.
- IFERROR is useful when a valid business output is needed instead of an Excel error.
- Criteria cells make formulas more reusable than hardcoding criteria repeatedly.
- Nested IF conditions should be ordered carefully from the strongest/most specific threshold to the weaker thresholds.
- Structured references such as [`@Sales`] improve readability when working with Excel Tables.

19. Day 03 Formula Cheat Sheet

`=IF(condition, value_if_true, value_if_false)`

`=IF(condition1, result1, IF(condition2, result2, result3))`

`=AND(condition1, condition2, ...)`

`=OR(condition1, condition2, ...)`

`=IFERROR(formula, value_if_error)`

`=SUMIF(criteria_range, criteria, sum_range)`

`=SUMIFS(sum_range, criteria_range1, criteria1, ...)`

`=COUNTIF(range, criteria)`

`=COUNTIFS(criteria_range1, criteria1, ...)`

`=AVERAGEIF(criteria_range, criteria, average_range)`

`=AVERAGEIFS(average_range, criteria_range1, criteria1, ...)`
